



KYNTRIX

— SIMULATION BEYOND LIMITS —



Kyntrix *Simulation Beyond Limits*

23AID202

Group B-14

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Literature Review :

Title	Author	Year	Venue	Outcomes	Limitations
WebGL-Based 3D GUI for Robot Modeling & Control	Ujjal Dey et al.	2020	CAIEE	• 3D robot modeling • FK & IK analysis • PUMA 560 control • CAD/STL integration	• Joint-range constraints • Some IK solutions infeasible
m-PaRoLa: Mobile Virtual Laboratory	Adrián Peidró et al.	2018	IFAC-PapersOnLine	• Mobile robot simulation • Visual kinematics • Five-bar & 3RRR robots	• JavaScript simulation only • Limited robot models
Web-Based Robotics Simulator	Amit Kumar Nandi et al.	2020	IC-RACT	• Robot modeling • FK & IK • Motion planning • Path planning	• Up to 6 DOF • Limited workspace mapping • Limited interoperability

Research Gaps :

1. Limited Robot Support :

Existing web-based simulators support only a limited number of robot models.

2. Separate Analysis Tools :

Important robotics calculations are often provided as separate tools instead of one integrated platform.

3. Computation–Simulation Gap :

There is a need to connect mathematical robot calculations directly with interactive 3D simulation.

4. Limited Practical Learning :

Limited support for combining simulation, analysis, and practical challenges in one web-based platform.

Objectives :

1. Unified Robot Simulation

Develop a web-based platform to simulate diverse industrial and educational robot models.

2. Integrated Robotics Analysis

Provide forward and inverse kinematics, Jacobian analysis, trajectory generation, and workspace analysis in one platform.

3. Real-Time 3D Visualization

Connect mathematical robot calculations with interactive 3D robot simulation.

4. Practical Robotics Learning

Provide structured challenges to help users apply robotics concepts through simulation.

Problem Statement :

1. **Fragmented Robotics Tools**

Students and researchers often need different tools for robot simulation, kinematic calculations, workspace analysis, and trajectory planning.

2. **Limited Integrated Platforms**

Existing web-based platforms have limited support for combining multiple robot models, robotics analysis, 3D simulation, and practical challenges in one environment.

Our Solution - KyntriX

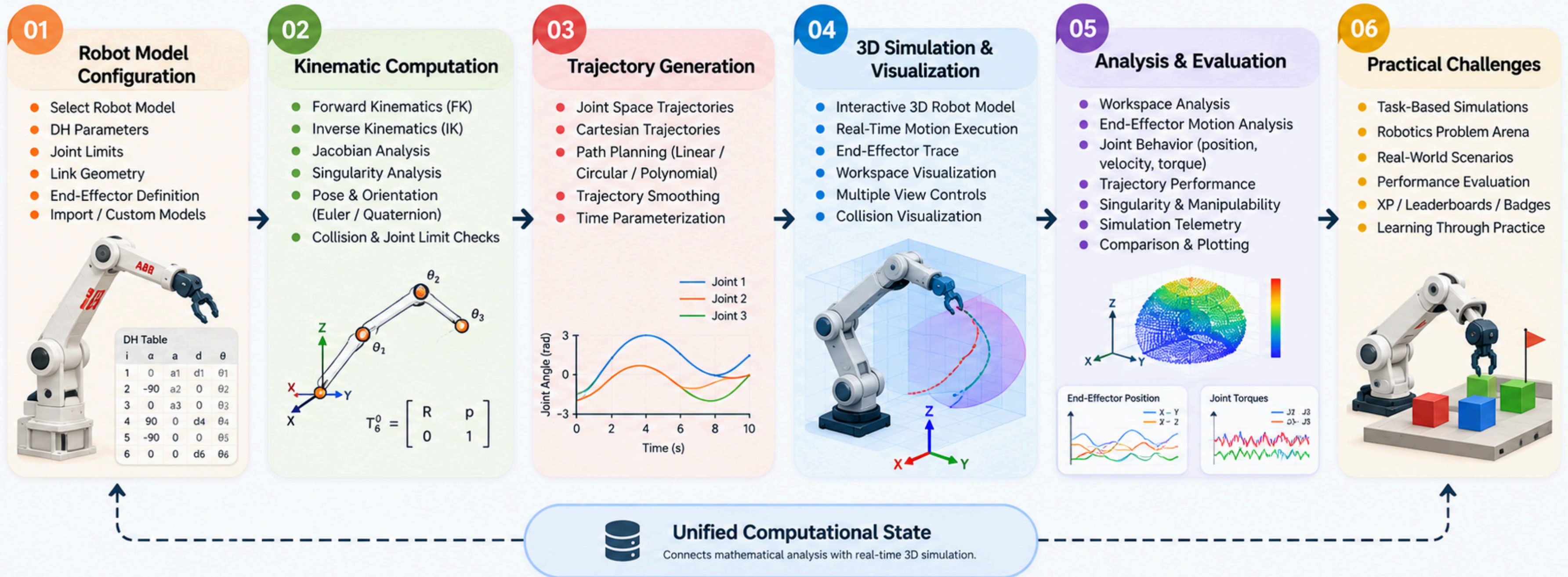
3. **Unified Web-Based Platform**

KyntriX brings robot simulation and robotics analysis together in a single browser-based environment.

4. **Integrated Robotics Environment**

It combines multiple robot models, kinematic analysis, interactive 3D simulation, and structured challenges in one platform.

Methodology :



Results :

1. Unified Robot Simulation

42 robot configurations supported in an interactive 3D environment.

2. Integrated Robotics Computation

FK, IK, Jacobian, trajectory, workspace and related analysis integrated into the simulator.

3. Working Web-Based Laboratory

A browser-based environment for simulation, analysis and practical robotics challenges.

**Robotic Manipulator Configuration Setup**
Select your industrial CAD robotic arm or theoretical academic skeleton to initialize live simulation

×

 Industrial (22)

 Skeletons (20)

 Custom Arm →

🔍 Search robot name or brand (Puma, KUKA, ABB, SCARA)...

DOF: **All** 1D 2D 3D 4D 5D 6D

Presets: Puma 560 (6R) KUKA KR5 ABB IRB 120 Stanford (5R1P) Planar (3R) SCARA (3R1P)

Limitations :

- No Physical Robot Integration
- Collision Detection Not Yet Implemented
- Workspace Boundary is Sampled.
- Browser Computation Has Practical Limits

Future Scope :

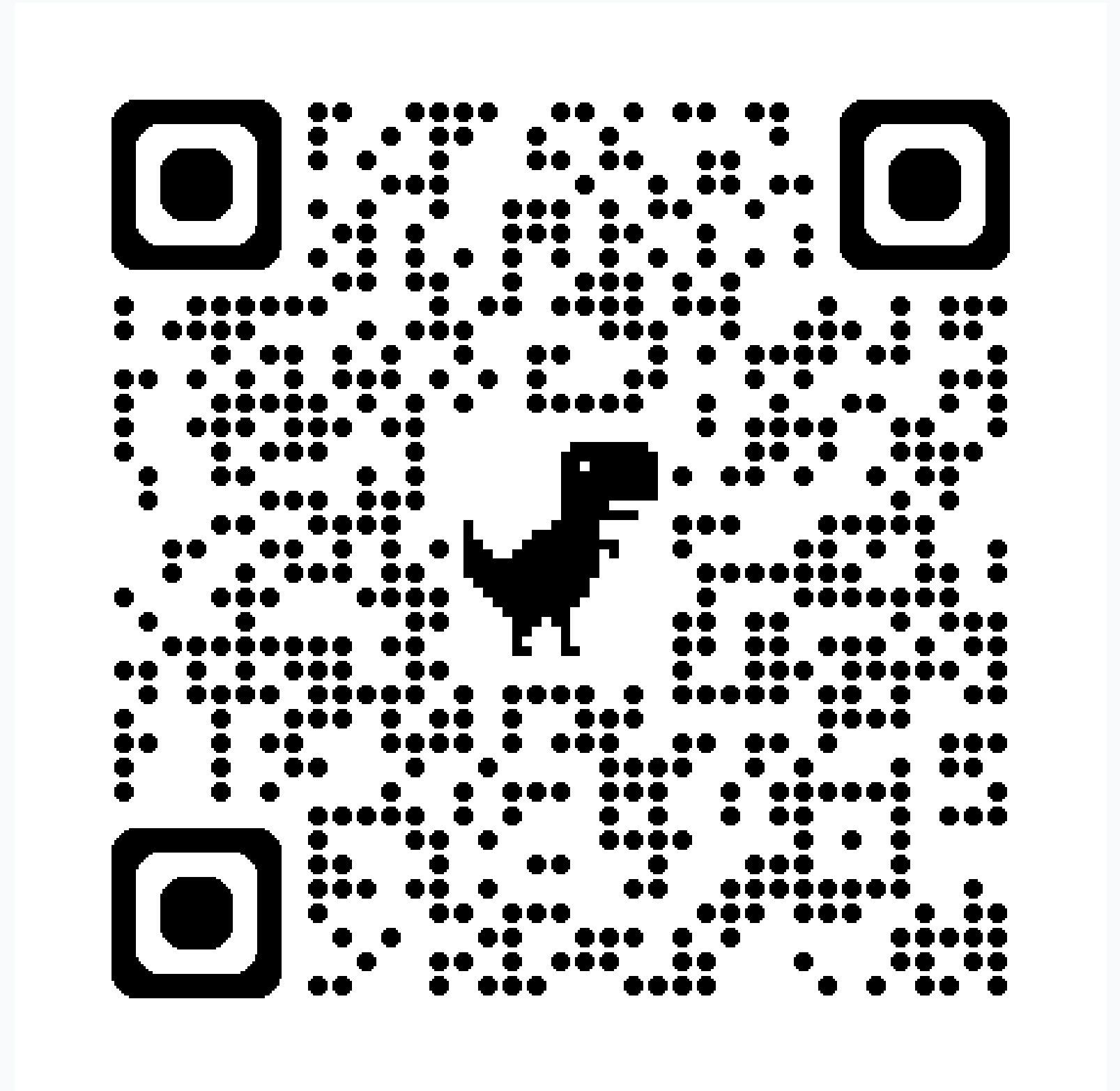
- Real-Robot Hardware Integration
- Add robot and environment collision detection
- Advanced Workspace Computation
- Improve computation speed and scalability

Conclusion :

- KyntriX brings robot simulation and kinematic analysis together in one platform.
- It supports different robot models with interactive 3D simulation and analysis.
- It provides a simple web-based environment for learning, testing, and experimenting with robotics.

URL - kyntrix.in.net

Google Search - **KyntriX Simulation**



References :

- Dey, U.; Cheruvu S., K. - Computer Applications in Engineering Education, 2020.

Link For the Website - <https://onlinelibrary.wiley.com/doi/10.1002/cae.22282>

Direct PDF Link - [Github Link](#)

- Peidr , A.; Tendero, C.; Mar n, J. M.; et al. - IFAC-PapersOnLine, 2018.

Link For the Website - <https://www.sciencedirect.com/science/article/pii/S2405896318303586>

Direct PDF Link - [Github Link](#)

- Nandi, A. K.; Pattanaik, L. N. - IC-RACT 2020, 2020.

Link For the Website - https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3697570

Direct PDF Link - [Github Link](#)

Testing Document for KyntriX - [**Click Me**](#)

Team Roles and Contributions:

Team Member -1 :

- Robot modelling and configuration
- DH parameter calculation
- Forward Kinematics
- Transformation matrices
- Robot position analysis

Team Member -2 :

- Inverse Kinematics
- Jacobian and velocity analysis
- Singularity detection
- Manipulability analysis
- Joint-limit handling

Team Member -3 :

- Joint motion planning
- Trajectory generation
- End-effector path planning
- Motion analysis
- Trajectory visualization

Team Member -4 :

- 3D robot simulation
- Robot visualization and controls
- Coordinate frames and end-effector tracking
- Robot model integration
- Integration of robotics modules

Thank
you!